Generalized linear models

Quasi-likelihood and generalized estimation equations

Prof. Dr. Andreas Groll

Winter semester 20/21

Quasi-likelihood and generalized estimation equations - Contents

Quasi-likelihood

- With help of the deviance analysis: Check discrepancy between model and data
- What to do if a model does not fit?
- Possible sources of error:
  1. Distribution model
  2. Response / link function
  3. Covariates and their structure

For a wrong distribution model: use quasi-likelihood!

Quasi-likelihood

Idea of quasi-likelihood

Likelihood procedure:

$b'(\mu_i)$

$\downarrow$

Assumption	Moments	Estimation
distribution + link function $\Rightarrow$	$E[y_i] = \mu_i$ $Var(y_i) = \phi v(\mu_i)$ $\Rightarrow$ $(v(\mu_i) = b''(\mu_i))$	$s(\beta) = \sum_{i=1}^{n} \frac{\partial h(\eta_i)}{\partial \eta} \frac{(y_i - \mu_i)}{\phi v(\mu_i)} \cdot \underline{x}_i$

e.g. Poisson-distr. + log-link

$E[y_i] = \mu_i = \exp(\underline{x}_i^T \beta) \quad (= \lambda_i)$

$Var(y_i) = \mu_i = \exp(\underline{x}_i^T \beta) \quad (= \lambda_i)$

Quasi-likelihood

Quasi-likelihood method:

- Assumption that the specification does not fit
- E.g. count data with larger dispersion than the model can account for (overdispersion) $(\mathbb{E}[y_i] \neq Var(y_i))$
- Suitable distribution?

Idea:
Start with the specification of $\mathbb{E}(y_i) = \mu_i = h(\eta_i)$ and $\text{Var}(y_i) = \phi v(\mu_i)$ without a specific distribution assumption, e.g. $\text{Var}(y_i) = \phi\mu_i$ for count data ("Poisson with dispersion").

new

Quasi-likelihood

- Instead of score function consider Quasi-score function:

$$\mathbf{s}_Q(\beta) = \sum_{i=1}^{n} \frac{\partial h(\eta_i)}{\partial \eta} \frac{(y_i - \mu_i)}{\phi v(\mu_i)} \mathbf{x}_i$$

working variance
is not necessarily
$= b''(\mu_i)$

- Both first moments are correctly specified via

$$\mathbb{E}(y_i) = \mu_i = h(\eta_i) \quad \text{and} \quad \mathrm{Var}(y_i) = \phi v(\mu_i)$$

- The quasi-likelihood is a function $Q(\cdot)$ with

$$\frac{\partial Q(\beta)}{\partial \beta} = \mathbf{s}_Q(\beta)$$

(we actually don't care about Q)

- The quasi-likelihood estimator approximately solves the estimation equation

$$\mathbf{s}_Q(\beta) = 0$$

Quasi-likelihood

- Estimation completely identical in quasi-likelihood and in actual likelihood case
- Conceptual differences:
  - for QL no distribution assumption
  - therefore, for QL any specification of variance function possible

Why then ever use the actual likelihood?

Quasi-likelihood

Asymptotics

Maximum likelihood:

$$\hat{\beta}_{\text{ML}} \overset{a}{\sim} N(\beta, \mathbf{F}(\beta)^{-1})$$

- consistent ✓ (asymptotically)
- unbiased ✓
- efficient ✓ → amongst all asympt. unbiased estimators our $\hat{\beta}_{ML}$ has the smallest variance

⇒ reaches the Cramér-Rao-boundary

Quasi-likelihood

Quasi-likelihood:

$$\hat{\beta}_{\text{QML}} \overset{a}{\sim} N(\beta, \mathbf{F}_Q(\beta)^{-1})$$

- consistent ✓
- unbiased ✓ (asympt.)
- but not necessarily efficient

$$\begin{aligned}
\mathbf{F}_Q(\beta) &= \mathbb{E}\left(-\frac{\partial \mathbf{s}_Q(\beta)}{\partial \beta}\right) \\
&= \mathbb{E}\left((\mathbf{s}_Q(\beta)\mathbf{s}_Q(\beta)^\top\right) \\
&= \mathbf{X}^\top \mathbf{D}(\beta) \mathbf{\Sigma}_Q^{-1}(\beta) \mathbf{D}(\beta) \mathbf{X}
\end{aligned}$$

Fisher regularity (at least partly) seems to hold

new; working covariance matrix

Example: Quasi-Poisson with $\text{Var}(y_i) = \phi \mu_i$:

$$\mathbf{F}_Q(\beta) = \frac{1}{\phi}\mathbf{F}(\beta)$$

$$\hat{\phi} = \frac{1}{n-p-1} \sum_{i=1}^{n} \frac{(y_i - \mu_i)^2}{\mu_i}$$

Generalized estimation equations (GEEs)

special case

1. Likelihood
   Distribution specified correctly, especially $\mathbb{E}(y)$ and $\mathrm{Var}(y)$

special case

2. Quasi-likelihood
   no distribution, but $\mathbb{E}(y)$ and $\mathrm{Var}(y)$ correctly specified
3. GEEs
   $\mathbb{E}(y)$ specified correctly, $\mathrm{Var}(y)$ not necessarily
   e.g. if there are dependencies between observations

Generalized estimation equations (GEEs)

Consider a model with

- correctly specified expectation $\mu_i = h(\mathbf{x}_i^\top \beta)$
- working covariance matrix $\sigma_{w,i}^2 = \phi v(\mu_i)$

For the true covariance usually holds: $\sigma_i^2 \neq \sigma_{w,i}^2$.

Obtain an estimator for β by (approximately) solving the generalized estimation equation:

$$\mathbf{s}_Q(\beta) = \sum_{i=1}^{n} \frac{\partial h(\eta_i)}{\partial \eta} \frac{(y_i - \mu_i)}{\phi v(\mu_i)} \mathbf{x}_i = \mathbf{X}^\top \mathbf{D}(\beta) \mathbf{\Sigma}_w^{-1}(\beta)(\mathbf{y} - \boldsymbol{\mu}) \overset{!}{=} 0$$

$$\Sigma_w = \phi \begin{pmatrix} v(\mu_1) & & 0 \\ & \ddots & \\ 0 & & v(\mu_n) \end{pmatrix}$$

Generalized estimation equations (GEEs)

How is such an estimator $\hat{\beta}$ asymptotically distributed?

- Taylor expansion of $\mathbf{s}_Q$ around $\hat{\beta}$

$$\underline{0} \overset{!}{=} \underline{s}_Q(\hat{\underline{\beta}}) = \underline{s}_Q(\underline{\beta}) + \frac{\partial \underline{s}_Q(\underline{\beta})}{\partial \underline{\beta}}(\hat{\underline{\beta}} - \underline{\beta}) + \dots$$

$$\Rightarrow \hat{\underline{\beta}} - \underline{\beta} \approx \left(-\frac{\partial \underline{s}_Q(\underline{\beta})}{\partial \underline{\beta}}\right)^{-1} \cdot \underline{s}_Q(\underline{\beta})$$

$$\mathrm{Cov}(\hat{\underline{\beta}} - \underline{\beta}) = \mathrm{Cov}\Bigg(\underbrace{\left(-\frac{\partial \underline{s}_Q(\underline{\beta})}{\partial \underline{\beta}}\right)^{-1}}_{\circledast} \cdot \underline{s}_Q(\underline{\beta})\Bigg) = \underline{\underline{A}}\ \mathrm{cov}(\underline{s}_Q) \cdot \underline{\underline{A}}$$

$\rightarrow \mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ if $\mathbb{E}(\mathbf{s}_Q(\boldsymbol{\beta})) = 0$ and

$$\rightarrow \mathrm{Cov}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \left(-\frac{\partial \mathbf{s}_Q(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right)^{-1} \mathrm{Cov}(\mathbf{s}_Q(\boldsymbol{\beta})) \left(-\frac{\partial \mathbf{s}_Q(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right)^{-1}$$

$\circledast =: \underline{\underline{A}}$, which is constant

Generalized estimation equations (GEEs)

(distribution assumption, mean, var correct)

In GLMs we have:
$\mathbf{s}_Q = \mathbf{s}$ and $-\frac{\partial \mathbf{s}_Q(\beta)}{\partial \beta} = \mathbf{F}_{\text{obs}}(\beta)$.
Moreover:

$$\text{Cov}(\mathbf{s}_Q(\beta)) = \mathbb{E}(\mathbf{F}_{\text{obs}}(\beta)) = \mathbf{F}(\beta)$$
$$\rightarrow \text{Cov}(\hat{\beta} - \beta) = \mathbf{F}_{\text{obs}}(\beta)^{-1}\mathbf{F}(\beta)\mathbf{F}_{\text{obs}}(\beta)^{-1}$$

For canonical link:

$\mathbf{F}_{\text{obs}}(\beta) = \mathbf{F}(\beta)$, so

$$\text{Cov}(\hat{\beta} - \beta) = \mathbf{F}(\beta)^{-1}$$

$$\text{Cov}(\hat{\beta}) = \text{Cov}(\hat{\beta} - \beta)$$

Generalized estimation equations (GEEs)

$=^{*}$ if working variance is equal to the true variance

GEEs:

$$\mathbb{E}\left(-\frac{\partial \mathbf{s}_Q(\beta)}{\partial \beta}\right) = \mathbf{F}_Q(\beta) \neq \mathrm{Cov}(\mathbf{s}_Q(\beta))$$

with $\mathbf{F}_Q(\beta) = \mathbf{X}^\top \mathbf{W}(\beta)\mathbf{X}$,

whereas

$$\mathbf{W}(\beta) = \begin{pmatrix} \left(\frac{\partial h(\eta_1)}{\partial \eta}\right)^2 \Big/ v(\mu_1)\cdot\phi & & \\ & \ddots & \\ & & \left(\frac{\partial h(\eta_n)}{\partial \eta}\right)^2 \Big/ v(\mu_n)\cdot\phi \end{pmatrix}$$

Altogether:

$$\mathrm{Cov}(\hat{\beta} - \beta) \approx \mathbf{F}_Q(\beta)^{-1} \underbrace{\mathrm{Cov}(\mathbf{s}_Q(\beta))}_{=:\underline{\underline{V}}(\beta)} \mathbf{F}_Q(\beta)^{-1}$$

Generalized estimation equations (GEEs)

To sum up:
Asymptotically under regularity conditions it holds that:

$$\hat{\beta} \overset{a}{\sim} \mathrm{N}\left(\beta, \mathbf{F}_Q(\beta)^{-1}\mathbf{V}(\beta)\mathbf{F}_Q(\beta)^{-1}\right)$$

For correctly specified variance: $\underline{\underline{V}}(\underline{\beta}) = \underline{\underline{F}}_Q(\underline{\beta})$

$$\hat{\beta} \overset{a}{\sim} \mathrm{N}\left(\beta, \mathbf{F}_Q(\beta)^{-1}\right)$$

In practice, for the computation of Wald and score test statistics:

- Replace $\mathbf{F}_Q(\beta)$ by $\mathbf{F}_Q(\hat{\beta})$ $\Rightarrow \hat{\mu}_i$
- Replace μ_i in $\mathbf{V}(\beta)$ by $h(\mathbf{x}_i^\top\hat{\beta})$.

$$\hat{\underline{\underline{V}}}(\hat{\underline{\beta}}) = \sum_{i=1}^{n} \underline{x}_i \underline{x}_i^\top \frac{\left(\partial h(\hat{\eta})/\partial\eta\right)^2}{\left(v(h(\underline{x}_i^\top\hat{\underline{\beta}}))\hat{\phi}\right)^2}\left(y_i - h(\underline{x}_i^\top\hat{\underline{\beta}})\right)^2$$